

CHEMICAL CHANGES – MAKING SOLUBLE SALTS

KEY VOCABULARY

Soluble salt: a salt that dissolves in water.

Excess: extra base added to use up all the acid.

Filtration: removes the unreacted excess base.

Crystallisation: evaporates water to leave salt crystals.

COMMON MISCONCEPTIONS

~~— Add just enough base to the acid.~~

✓ Add base in EXCESS so all the acid reacts.

~~— Boil the solution dry to get crystals.~~

✓ Evaporate gently and let it crystallise.

~~— Filter to remove the salt.~~

✓ Filter to remove the unreacted excess solid.

METHOD (INSOLUBLE BASE)

Warm the acid in a beaker.

Add the base (e.g. copper oxide) until in excess.

Filter to remove unreacted base.

Heat the filtrate to the point of crystallisation.

Leave to crystallise, then pat dry.

1 WHY EXCESS

Explain why the base is added in excess. [2 marks]

2 REMOVE EXCESS

Name the technique used to remove the unreacted base. [1 mark]

3 GET CRYSTALS

Describe how to obtain dry crystals from the salt solution. [3 marks]

4 CHOOSE REACTANTS

Name the acid and base needed to make copper sulfate. [2 marks]

5 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

Add the base in ____, then ____ off the unreacted solid.

excess

filter

evaporate

boil

6 WHY GENTLY

Explain why the solution is not heated strongly to dryness. [1 mark]
